

# Distance between two lines

## Key Points

The distance between parallel lines can be worked out by picking a point on one.

For skew lines

$$r_1 = a_1 + td_1 \text{ and } r_2 = a_2 + td_2$$

The distance is

$$\frac{|(a_2 - a_1) \cdot n|}{|n|}$$

Where the normal  $n = d_1 \times d_2$

## Basic Skills

**Q1.** Define what is meant by skew lines

**Q2.** Find the distance between the point  $(6, 2, -1)$  and  $(2, 3, 2)$

**Q3.** For the line with point  $(2, 1, 2)$  and a direction of  $d = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$

a) Write down line in the form  $r = a + \lambda d$

b) Write down line in the form  $(r - a) \times b = 0$

**Q4.** For  $r_1 = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} + t \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$  and  $r_2 = \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} + s \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$  determine any points of intersection

## Competency Skills

**Q5.** For the lines  $r_1 = \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$  and  $r_2 = \begin{pmatrix} 5 \\ -1 \\ -2 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$

a) Find the normal between the directions of the lines

c) Find the distance between the lines by using  $\frac{|(a_2 - a_1) \cdot n|}{|n|}$

**Q6.** Find the distance between lines  $r_1 = \begin{pmatrix} -5 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$  and  $r_2 = \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$

**Q7.** Find the distance between lines  $r_1 = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix}$  and  $r_2 = \begin{pmatrix} 0 \\ 1 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} -12 \\ 6 \\ -30 \end{pmatrix}$

## Advanced Skills

**Q8.** For coordinates  $A=(0,1,0)$ ,  $B=(5,3,2)$ ,  $C=(-1,2,3)$  and  $D=(4,2,-4)$

a) Find the equation for the line AB and CD

b) Find the vector product of  $\overrightarrow{AB} \times \overrightarrow{CD}$

c) Find the distance between the lines AB and CD using the formula

**Q9.** Find the shortest distance between the line of intersection of the planes  $3x - y + z = 2$  and  $2z - x = 5$  and the line  $y = 3x, z = 3$

**Q10.** Explain where the formula comes from

## Extension

**Q10.** Predict how planes will interact in 4D

Hint: Consider how lines interact in 3D